
Bi-GRUBERTa model for Dialogue Act Classification

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Abstract

We introduce Bi-GRUBERTa, a novel hybrid model for dialogue act classification that combines transformer-based embeddings with recurrent decoding. Our approach fine-tunes RoBERTa to produce turn-aware utterance representations, enhanced by a binary speaker-turn feature and contextualized through a bidirectional GRU. These enriched embeddings are then decoded by a unidirectional GRU with hard guided attention, preserving dialogue order and aligning predictions with their corresponding utterances. In empirical evaluations, Bi-GRUBERTa slightly outperforms a retrained version of the model proposed by He et al. [2021] across both conversational benchmarks. Our evaluation shows that integrating guided attention into a recurrent decoder can effectively enhance modern pretrained encoders by enriching contextual representations and improving label prediction with minimal computational overhead. We conclude by exploring the impact of model design choices on label imbalance and outlining opportunities for future enhancements, including dialogue-specific pretraining and sequence-level fine-tuning objectives.

1 Introduction

In recent years, the development of intelligent conversational systems has significantly advanced, driven by the increasing demand for human-computer interaction in domains such as virtual assistants, customer support, and dialogue-based tutoring systems. A fundamental component enabling these systems to interpret and respond appropriately to user input is Dialogue Act (DA) classification, which involves identifying the communicative intent behind each utterance in a conversation (Ahmadvand et al. [2020]). By assigning a DA label—such as Statement, Question, or Backchannel—to each utterance, DA classification supports tasks in Natural Language Understanding (NLU) and Natural Language Processing (NLP) (Ahmadvand et al. [2020], Shang et al. [2020]).

The following (Table 1) is a dialogue comprising seven utterances exchanged between two speakers. Each utterance is annotated with a corresponding dialogue act label, such as Question or Backchannel, reflecting its communicative function within the interaction (Raheja and Tetreault [2019]):

Speaker	Utterance	DA label
A	Okay.	Other
A	Um, what did you do this weekend?	Question
B	Well, uh, pretty much spent most of my time in the yard.	Statement
B	[Throat Clearing]	Non Verbal
A	Uh-Huh.	Backchannel
A	What do you have planned for your yard?	Question
B	Well, we're in the process of revitalizing it.	Statement

Table 1: *Example of a dialogue with Dialogue Act (DA) labels from SwDA corpus.*

1.1 Motivations and Contributions: Bridging the Gap Between Speaker-Aware and Turn-Aware Models

Dialogue Act classification is essential to comprehend and interpret conversations properly to improve the performances of NLP and NLU systems. However, this task still poses significant challenges, because state-of-the-art models focus either on complex hierarchical architectures that encode speaker identity Colombo et al. [2020] or on simplified embeddings that incorporate only turn transitions He et al. [2021] without integrating the strengths of both paradigms. Motivated by these limitations, we propose a hybrid model that leverages the strengths of both approaches. Specifically, we combine RoBERTa-based utterance embeddings with speaker turn-aware information and incorporate a GRU-based decoder with Hard Guided Attention, adopting a sequence-to-sequence (Seq2Seq) framework. This allows us to capture both the local contextual dependencies and the broader structure of dialogue interactions, and our approach will be evaluated on standard datasets (SwDA and MRDA), demonstrating competitive performance and highlighting the benefits of our hybrid model.

1.2 Datasets

As previously stated, we evaluate the performance of our model on two datasets: the Switchboard Dialogue Act Corpus (SwDA) University [2023], Jurafsky [1997] and the Meeting Recorder Dialogue Act Corpus (MRDA) Dhillon et al. [2004], Shriberg et al. [2004].

SwDA: The SwDA corpus contains 1,155 telephone conversations, each lasting approximately five minutes, involving 440 speakers and totaling over 221k utterances. The conversations revolve around predefined topics such as childcare and recycling. Utterances are annotated using SWBD-DAMSL, a refined version of the original DAMSL tagset, organized into 220 dialogue act (DA) labels grouped under 42 broader classes. The annotation schema is multi-dimensional, including aspects such as communication status, information level, forward and backward functions. The annotation process involved multiple annotators, with an inter-annotator agreement of 0.80 (Cohen’s Kappa).

MRDA: The MRDA dataset includes over 180k DA annotations from 75 real-world meetings (approximately 72 hours of speech) recorded at the International Computer Science Institute. It features 53 speakers and uses a modified version of SWBD-DAMSL to better reflect the meeting setting. There are 3 different levels of dialogue act labels. The most generic one has 52 total labels, the general one has 12, and the basic one has just 5. The corpus includes three types of annotations: utterance boundaries, DA labels (with general and specific tags), and adjacency pairs (e.g., question-answer). Like SWDA, MRDA was labeled by multiple annotators, yielding a Kappa score of 0.80.

2 Background - State of The Art

Before talking about our selected approach we are briefly going to discuss two of the approaches that reach some the highest results on the two introduced datasets. Both papers present approaches that try to exploit speaker/turn information, which is what differentiates plain written text from dialogue. They have two different approaches, one focusing on the speaker identity and the other focusing only on the turn taking aspect, ignoring identity. The previous approaches didn’t consider any of this since at the time since most of the other approaches relied on the use of linear Conditional Random Field (CRF) combined with a recurrent neural network based encoder (Li et al. [2018], Chen et al. [2017]).

2.1 Guiding Attention in Sequence-to-Sequence Models for Dialogue Act prediction

In this first paper by Colombo et al. [2020] they treat the DA Classification task as a Seq2Seq one. The idea behind this approach is to use multiple levels of embedding through the use of a Hierarchical Encoder (HGRU) and then pass them to a decoder with attention to get the output that needs to be classified. The encoder is made of three layers of Bi-GRU networks (Chen et al. [2017]) each creating a different embedding that depends on the previous one. The three levels are: Word level, which creates an embedding for each word of the input utterances, Persona level which encodes speaker identity information in the Word embedding and finally the Sentence level which creates the final embedding, one per utterance. The last hidden state of the last layer of the encoder becomes the first hidden state of the decoder. This decoder is a GRU network with the addition of a novel form of Attention, called Guided Attention. The idea is to exploit the perfectly aligned task to 'guide' the values of the $\alpha_{j,k}$ by simply adding 1 to the corresponding timestep in the case of Soft Guided Attention or by forcefully make it be the only non-zero value in the Hard Guided Attention. The result of this attention is a context vector c_k that is used alongside the hidden state and the previously predicted label to decode the utterance. During training they also used teacher forcing to replace the last predicted label with the true one. This architecture produce very good results, achieving state of the art on the SwDa dataset with 85.0% accuracy and very close to state of the art for MRDA with 91.6%. They also note that this way of encoding speaker information may be more efficient on two-people conversations rather than multi-party ones considering the results.

2.2 Speaker Turn Modeling for Dialogue Act Classification

In this second paper published by He et al. [2021] they step away from the Seq2Seq approach, and instead put their focus on finding a simpler and more efficient architecture than the previous one, leveraging speaker turn information, not considering their identity. In this case the encoding is done by combining an embedding produced by a pretrained Language Model, one per utterance, and we add a 0/1 value called Speaker Turn Embedding that is determined like this:

$$turn(i, j) = \begin{cases} 1 & \text{if } speaker_i \neq speaker_j \\ 0 & \text{if } speaker_i = speaker_j \end{cases}$$

In particular, in the paper they choose RoBERTa (Liu et al. [2019]), as the pretrained Language Model to compute the embeddings. The combined embedding, which they call "speaker turn-aware utterance embeddings" are then fed to a Bi-GRU to produce a contextualized version. Unlike in the first paper, this approach does not decode the final embeddings, but instead classifies directly on them with a fully connected layer with a softmax on top. They proved that even with a transformer-based approach, they could still reach good results on both datasets, 83.2% on SwDa and 91.4% on MRDA, also suggesting that this approach to encode turn information, may be more successful for multi-party conversations.

3 Method

The idea behind our implementation is to combine the two discussed approaches, in hopes of getting the best of both worlds. So following the formulation of Colombo et al. [2020], Dialogue-Act (DA) classification is modeled as a *sequence-to-sequence labeling* problem. Given a dialogue $\mathcal{D} = \langle u_1, \dots, u_n \rangle$ of n utterances, the model produces a label sequence $\langle y_1, \dots, y_n \rangle \in \mathcal{Y}^n$ where each y_i characterizes the communicative function of u_i (e.g. STATEMENT, QUESTION). Treating the dialogue as a time-ordered sequence allows the decoder to leverage contextual dependencies, for example allowing to increased the likelihood of an having an ANSWER given a QUESTION, and leveraging the alignment of the task in the *Guided Attention* mechanism

Unlike the original guided-attention architecture, which stacks multiple hierarchical GRU layers to encode words, personas and sentences, we adopt the transformer based backbone proposed by He et al. [2021]. Each utterance is embedded once with a fine-tuned RoBERTa encoder and augmented by a binary *speaker-turn* embedding that flags whether the current speaker differs from the previous one. This transformer-based, turn-aware sequence is then fed to our decoder with *hard guided attention*. The comparison aims to test whether modern transformer utterance embeddings plus minimal turn signaling can outperform a deeper, word- and persona-level GRU stack while preserving the original sequence-labeling philosophy.

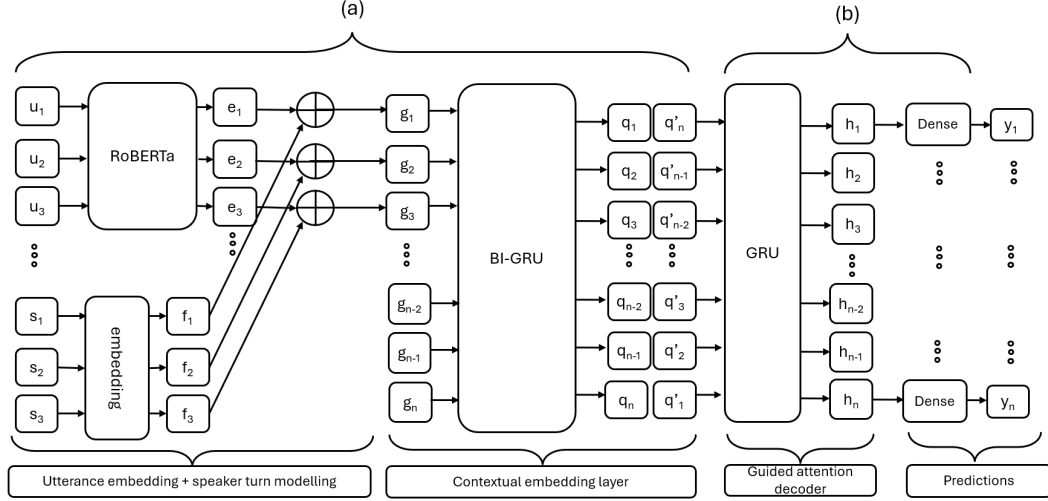


Figure 1: Visual representation of the Bi-GRUBERTa architecture (a) The encoder, composed by RoBERTa and a speaker-turn aware embedding plus a Bi-GRU to enrich the embeddings with context. (b) The GRU decoder that leverages guided attention.

Our training pipeline initially closely followed the one of He et al. [2021], however, early-stage experiments exposed pronounced overfitting well before the 50-epoch convergence they report. To mitigate this, we adopted a progressive unfreezing protocol inspired by Howard and Ruder [2018], incrementally unlocking RoBERTa’s layers to shield its pretrained representations from catastrophic forgetting while enabling our classification head to adapt without being overwhelmed in the initial training phases.

4 Experimental analysis

4.1 Task(s)

Learning problem: Given a dialogue $\mathcal{D} = \langle u_1, \dots, u_n \rangle$ we learn a mapping $f_\theta : \mathcal{D} \rightarrow \mathcal{Y}^n$, $\mathcal{Y} = |\text{labels}|$. The model emits one distribution per utterance, $P_\theta(y_i | u_{\leq i})$.

Objective function: Training minimizes *token-level cross-entropy* with teacher-forcing:

$$\mathcal{L}(\theta) = - \sum_{i=1}^n \log P_\theta(y_i^* | u_{\leq i}, y_{< i}^*),$$

where y_i^* is the ground-truth DA for utterance u_i . Class imbalance is counteracted by inverse-frequency re-weighting ($w_c = 1/\sqrt{\text{freq}(c)}$), a compromise that stabilizes minority classes without inflating variance.

We also experimented fine-tuning the model by minimizing the expected risk loss ($\mathcal{L}_{\text{RISK}}$) inspired by Wiseman and Rush (2016), defined as:

$$\mathcal{L}_{\text{RISK}} = \frac{\sum_{\tilde{y} \in U(u_i)} \text{cost}(\tilde{y}, y_i) p(\tilde{y} | u_i)}{\sum_{\tilde{y}' \in U(u_i)} p(\tilde{y}' | u_i)} \quad (1)$$

where $U(u_i)$ is the set of sequences generated by the model for input u_i using a beam search algorithm, and the cost function is defined as

$$\text{cost}(\tilde{y}, y_i) = \begin{cases} 1 & \text{if } \tilde{y}_i \neq y_i \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

However, we found that applying this loss function led to excessive penalization, resulting in suboptimal performance. As a result, we opted not to adopt this approach in our final setup. Instead, we scheduled the teacher forcing ratio during training, gradually reducing it up to a minimum of 0.5.

Dataset breakdown: For **SwDA**, we start by taking the original 43 labels (see Table 2), but later reduce this number to 41. Utterances marked as non-verbal ('x' tags; e.g., 'Laughter', 'Throat_clearing') are removed and so is the label. Some utterances are labeled with the '+' tag, meaning that the speaker was interrupted, and the actual tag of the utterance is the one of the following utterance pronounced by the same speaker. We removed this label from the list and instead concatenated the discontinued sentences, to form a single uninterrupted sentence. All disfluency annotations (e.g., #, <, >, etc.) are removed to avoid leaking acoustic cues into the text-only model. This way, the total number of labels becomes 41.

Table 2: SwDA top-10 label distribution (covers 77 % of the corpus).

Label	Total		Train		Dev		Test	
	Cnt.	%	Cnt.	%	Cnt.	%	Cnt.	%
Statement-non-opinion (sd)	75 136	37.62	72 549	37.71	1 270	38.81	1 317	32.30
Backchannel (b)	38 281	19.17	36 950	19.21	567	17.33	764	18.73
Statement-opinion (sv)	26 421	13.23	25 087	13.04	616	18.83	718	17.61
Conventional-closing (fc)	2 581	1.29	2 480	1.29	20	0.61	81	1.99
Wh-Question (qw)	1 976	0.99	1 896	0.99	25	0.76	55	1.35
Response-acknowledgement (bk)	1 306	0.65	1 271	0.66	7	0.21	28	0.69
Hedge (h)	1 226	0.61	1 181	0.61	22	0.67	23	0.56
Open-Question (qo)	656	0.33	631	0.33	9	0.28	16	0.39
Other Answers (no)	284	0.14	277	0.14	1	0.03	6	0.15
Thanking (ft)	78	0.04	67	0.03	4	0.12	7	0.17

For **MRDA**, out of the three possible labeling levels, we adopt the more collapsed one with just five labels S, B, D, F, Q (see Table 2), following the literature’s approach. To remove noisy data, non-verbal (x tag) and non-labeled (z tag) utterances are deleted and disfluencies are handled the same way we did for SwDA.

Table 3: MRDA *Basic* 5-way label distribution across the official splits.

Label	Total		Train		Dev		Test	
	Cnt.	%	Cnt.	%	Cnt.	%	Cnt.	%
Statement (S)	64 233	59.36	45 099	60.08	9 563	58.19	9 571	57.30
Backchannel (B)	14 620	13.51	10 265	13.67	2 203	13.41	2 152	12.88
Disruption (D)	14 548	13.45	9 739	12.97	2 470	15.03	2 339	14.00
FloorGrabber (F)	7 818	7.23	5 324	7.09	1 085	6.60	1 409	8.44
Question (Q)	6 983	6.45	4 640	6.18	1 112	6.77	1 231	7.37

Official splits: We use the canonical *train/dev/test* partitions to maintain comparability. MRDA: 51/12/12 meetings (75k / 16k / 16k utterances). SwDA: 1115/21/19 calls (192k / 3k / 4k utterances). Note that our validation split is not identical to that used in He et al. [2021]; however, given that many prior works employ varying data splits for the validation, we opted to follow the original dataset partitioning for consistency and comparability with the broader literature.

Pre-processing pipeline:

Step 1 – Dialogue-specific cleaning Both datasets are pre-processed as previously described, including removal of non-verbal and non-labeled utterances, concatenation of interrupted segments, and stripping of disfluency annotations. We found an already preprocessed version of both datasets done by the same person <https://github.com/NathanDuran>, that was handled how we intended to, so we used it as a starting base.

Step 2 – Conversation indexing Each utterance is assigned a unique conversation identifier column, ensuring all utterances can be grouped by the dialogue they belong to.

Step 3 – Speaker-change flagging A binary column is added to each utterance following the method of He et al. [2021], where the flag is set as $\text{turn}(u_i) = 1[\text{speaker}(u_i) \neq \text{speaker}(u_{i-1})]$ to indicate speaker changes within the conversation.

Step 4 – Chunking Utterances are grouped by conversation and partitioned into fixed-length windows (chunks) of up to k turns. Each chunk aggregates text, speaker-change flags, dialogue act labels, and conversation IDs as parallel lists.

Step 5 – Batching and padding We implement a custom batching function (`collate_fn`) that pads each chunk at the utterance level to a uniform length (k turns per chunk). All utterances in a batch are then flattened and tokenized in parallel using the RoBERTa tokenizer, and the resulting tensors are reshaped to (batch size $\times k \times$ sequence length). The collate function also generates utterance masks, padding masks, and zero-padded label arrays, ensuring that speaker-change and label targets are aligned for efficient batch processing.

Evaluation protocol: To ensure fair and meaningful comparison, we experimented with various hyperparameter settings drawn from prior literature, running both a baseline model (replicating He et al. [2021]) and our proposed approach on the same dataset splits and with the same techniques. Due to computational resource constraints, we did not perform multiple repetitions or report average performance over different seeds, instead, each model was trained and evaluated once on the prescribed split.

Evaluation metrics: For our model, validation loss was computed under teacher forcing to mirror the training regime and used it to monitor learning progress and prevent overfitting. Following standard practice and He et al. [2021], we selected the checkpoint with the highest validation accuracy for final evaluation on the test set. Test performance is assessed using both overall accuracy and macro-averaged F1 score, the latter to fairly assess model performance across all labels since the dataset is imbalanced. Note that, unlike our model, the baseline does not employ a decoder, so teacher-forcing validation loss is not applicable in that setting.

4.2 Experimental settings

Hyperparameters: Both models were evaluated with a learning rate of 1×10^{-4} , consistent with prior work He et al. [2021]. We also evaluated 1×10^{-5} , which is the default for RoBERTa, to assess whether a more conservative learning rate would yield greater stability and finer-grained adaptation when fine-tuning a large pretrained model. Additionally, we implemented a hybrid learning rate schedule: when the RoBERTa encoder was frozen, we used a higher learning rate of 1×10^{-4} to facilitate faster adaptation of the task-specific BiGRU and GRU decoder layers but when the RoBERTa encoder was unfrozen, we reduced the learning rate to 1×10^{-5} to avoid overwriting useful pretrained representations and to ensure that updates to the backbone were more gradual.

Most parameter values, including the optimizer and overall training procedure, were based on He et al. [2021]. For aspects not specified in their work, such as weight decay and the hidden dimension for the BiGRU and GRU decoder, we adopted the configuration from Colombo et al. [2020].

We set the hidden sizes for BiGRU and GRU decoders to 128, as suggested by the literature, and subsequently experimented with a larger hidden dimension (256) to determine if increased capacity would improve performance.

In preliminary experiments, we also tested the original chunk sizes used in previous work, specifically 128 for SwDA and 350 for MRDA, but encountered significant GPU memory limitations, particularly for MRDA which forced us to reduce the batch size to 1 instead of 2. As a compromise, we adopted a chunk size of 64 for SwDA and 175 for MRDA, although the latter yielded the lowest performance among the configurations tested. This approach ensured our experimental setup was both well-founded and comparable to prior work, while remaining feasible within our hardware constraints. All the models were trained using Tesla T4 GPU with 16GB of RAM from both Google Colab and Kaggle.

Table 4: Hyperparameter settings for SwDA and MRDA experiments.

Hyperparameter	SwDA	MRDA
Learning rate	1×10^{-4} , 1×10^{-5} , hybrid	1×10^{-4} , 1×10^{-5} , hybrid
Weight decay	1×10^{-5}	5×10^{-5}
Gradient clip norm	5	5
Chunk size	64	175
Hidden size (Bi-GRU/GRU)	128, 256	128, 256
Optimizer	AdamW	AdamW

Freezing and unfreezing strategy: Following the Universal Language Model Fine-tuning (ULMFiT) methodology [Howard and Ruder, 2018], we initially froze all RoBERTa encoder parameters to prevent the large backbone from impeding the adaptation and stabilization of the Bi-GRU contextual encoder and GRU decoder. For the first two epochs, only the upper layers were trainable. We then incrementally unfroze three encoder layers per epoch, so that by the fourth unfreezing step, the entire RoBERTa model, including both token and speaker-turn embeddings, was trainable. Training with all layers unfrozen continued until an increase in validation loss was observed, at which point all but the top layers were refrozen for final fine-tuning. During this phase, the teacher forcing ratio was linearly decayed by 0.05 per epoch until reaching a minimum of 0.5. This gradual unfreezing protocol, motivated by transfer learning theory, is intended to mitigate catastrophic forgetting and support stable adaptation of both backbone and task-specific layers. We hypothesized that this approach would be particularly beneficial for our model architecture, which introduces an additional layer of complexity compared to the baseline, by allowing the task-specific upper layers to adapt more effectively before fully fine-tuning the entire network.

This strategy was initially implemented during early-stage experimentation to determine whether its application could be sustained throughout subsequent hyperparameter tuning phases and leveraged for efficiency gains. We evaluated its impact on both our Bi-GRUBERTa implementation and the He et al. baseline model. The results demonstrated tangible improvements in training efficiency: the gradual unfreezing approach delivered a 4.5% reduction in total training time for Bi-GRUBERTa and an 11.5% reduction for the baseline model. The per-epoch efficiency gains were even more pronounced, with average epoch times sped up by $1.40\times$ for Bi-GRUBERTa and $1.25\times$ for the baseline model.

Notably, these gains did not come at the expense of validation performance. In both models, validation loss trajectories remained consistently lower with gradual unfreezing, while training losses were comparable across both regimes, as illustrated in B.

Beam Search decoding: Instead of greedily selecting the single most probable token at each decoding step, we employ beam search with a beam width of k , maintaining the top- k candidate sequences based on their cumulative log-probabilities. At each decoding timestep, all active sequences are expanded by all possible next tokens, and only the top- k sequences (ranked by accumulated score) are retained. This process continues until all sequences reach an end-of-sequence token or a predefined maximum length. The final predicted utterance is selected as the sequence with the highest overall score, leading to more globally coherent outputs compared to greedy decoding.

Empirically, increasing the beam size from 1 to 5 we noticed a marginal improvement in test accuracy, ranging from approximately 0.1% to 0.4%, obtaining consistent performance boost as reported by Colombo et al. [2020]. However, this modest gain comes at a computational cost: on average, inference time increases by a factor of $\sim 1.90\times$ to $\sim 2.40\times$, compared to beam size 1.

Model selection and early stopping: Early stopping was employed with a patience of 3 epochs during fine-tuning and 2 epochs while the RoBERTa encoder was unfrozen, with model weights restored to those with the lowest validation loss. The final model for evaluation was selected based on the highest validation accuracy on the validation split, following He et al. [2021]. While the referenced paper trains for 50 epochs, we set a maximum of 20 epochs due to resource constraints; models with higher learning rates typically converged in fewer epochs. No repetitions or seed averaging were performed due to computational limitations.

4.3 Results

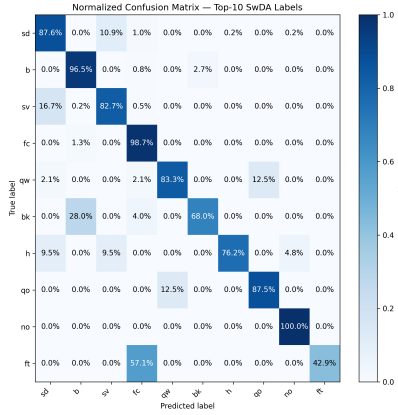
As we previously anticipated, we use as a baseline a reproduction of the model proposed by He et al., trained using the same techniques that we use for our model such as gradual unfreezing of RoBERTa, hybrid learning rate and 256-dimensional hidden size for the GRUs. Below are reported the accuracy results in Table 5 and the total training times in Table 6, along side the confusion matrices for both SwDA (Figure 2a) and MRDA (Figure 2b) datasets only for our model (the ones for the baseline can be found in Appendix A. In addition to accuracy, since both datasets are highly unbalanced, we also consider F1 macro average score in Table 7 to analyze the difference in performance for both models. Values for the state of the art models are missing since they are not provided.

Dataset	SwDA	MRDA
He et al. baseline (256)	78.45	90.26
Bi-GRUBERTa (256)	78.72	90.70
Seq2Seq (2020)	85.0	91.6
He et al. (2021)	83.2	91.4

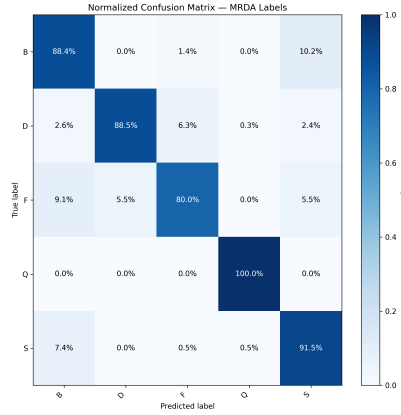
Table 5: Accuracy values for both datasets using hybrid learning rate and 256-dimensional embedding. On the bottom part are reported the original results for both papers

Model	Training time	Epochs
He et al. baseline (256)	4h 38m	15
Bi-GRUBERTa (256)	6h 3m	19

Table 6: Total training time and number of epochs for both architectures, different epoch value due to early stopping



(a) Bi-GRUBERTa confusion matrix for top-10 SwDA labels



(b) Bi-GRUBERTa confusion matrix for MRDA labels

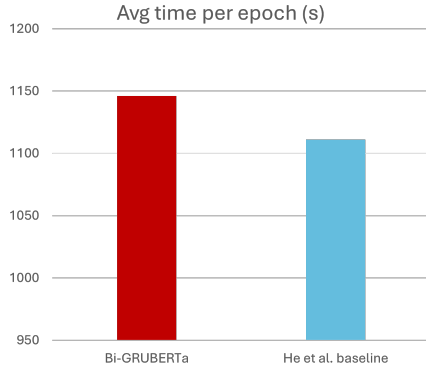
Dataset	SwDA	MRDA
He et al. baseline (256)	0.56	0.88
Bi-GRUBERTa (256)	0.57	0.89

Table 7: F1 score values for both datasets using hybrid learning rate and 256-dimensional embedding

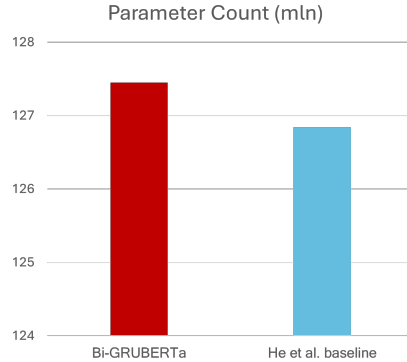
5 Discussion

Overall as shown from the results in Table 5 and Table 7, Bi-GRUBERTa achieves a small improvement in both accuracy and F1 score for both datasets. That being said, due to hardware limitations, we were unable to run multiple training runs for both models, so we cannot ensure that these values are statistically significant. If the improvements turn out to be significant, it is interesting to note that

the parameter difference between Bi-GRUBERTa and He et al. implementation is marginal (~ 0.6 M) as we can see from Plot 3b, resulting only in a slight increase in average training time per epoch (Plot 3a). This makes Bi-GRUBERTa a reasonable alternative to the baseline.



(a) Average training time per epoch of best configurations



(b) Parameter count of best configurations

Our model achieve high accuracy on SwDA, but the macro F1 score remains low, just above 0.5 for both models. The confusion matrix 2a, using the same method as Colombo et al. [2020], displays only the ten most frequent dialogue acts, which account for 77% of all instances among the 41 possible labels. This demonstrates that the models perform well on dominant classes but fail to generalize to rare, low-frequency categories. Published results typically report only accuracy and omit macro F1, we assume this result is common in their studies, and directly reflects the class imbalance inherent in dialogue-act classification.

In contrast, results on the MRDA dataset present a different scenario. As shown in the normalized confusion matrix (Figure 2b), the model achieves consistently strong performance across all five MRDA labels. This is further supported by the metrics in Tables 5 and 7, where both macro F1 and overall accuracy for MRDA are substantially higher than for SwDA. For example, our Bi-GRUBERTa model attains a macro F1 of 0.89 and an accuracy of 90.70% on MRDA, outperforming slightly the baseline model and demonstrating balanced generalization across classes. While the class distribution in MRDA remains imbalanced, the smaller label set and dataset size allow the model to achieve robust and consistent performance on both frequent and less frequent categories.

We also note the lack of transparency in the literature regarding the specific preprocessing steps applied by the baseline papers. In light of this, we adopted the preprocessing method proposed in <https://github.com/NathanDuran>, as detailed in the Task section 4.1. This approach standardizes both the SwDA and MRDA datasets, but for SwDA, it reduces the number of dialogue-act labels to 41 rather than the original 43. This reduction can significantly impact accuracy and F1 scores, potentially explaining the larger discrepancies we observe between our SwDA results and those reported in prior work. In contrast, the effect is less pronounced for MRDA, resulting in a smaller performance gap. For this reason, we chose to implemented a direct replication of the He et al. model to ensure a fair and consistent comparison with our own approach. Moreover, we diverge from both baseline papers in our choice of dataset split. Notably, each of the reference works applies a different partitioning strategy, so our approach retains the original dataset split of the dataset, resulting in a substantially smaller validation set (approximately 3,000 examples, compared to the 25,000 and 20,000 validation sizes reported in the literature). Furthermore, our chosen split for SwDA includes six missing dialogue-act labels, which may further compromise comparability and potentially affect the reliability of our reported results.

6 Conclusions

In this work, we introduced Bi-GRUBERTa, a hybrid Sequence-to-Sequence model that combines fine-tuned RoBERTa utterance embeddings with a GRU decoder enhanced by hard guided attention and simple speaker-turn embeddings. The goal of this project was to see if combining the approaches by Colombo et al. [2020] and He et al. [2021] we could improve the performance of He et al. model, which does not explore the possibility of using a more complex decoder, and instead only focuses on efficiently exploiting turn information.

To determine this, we evaluated our model on both the SwDA and MRDA corpora, obtaining test accuracies of 78.72% and 90.36%, respectively, against a retrained version of He et al. baseline, which obtained 78.45% and 89.29% test accuracy. Training Bi-GRUBERTa required approximately 6h 03m and introduced an additional ~ 0.6 million parameters, compared to the 4h 37m needed for the baseline. After a model selection phase, in which we tested different hyperparameter values, we found that the best configuration was hybrid learning rate schedule ($1 \times 10^{-4} \rightarrow 1 \times 10^{-5} \rightarrow 1 \times 10^{-4}$), 256 hidden-dimensional GRU embeddings, chunk sizes of 64 (SwDA) and 175 (MRDA), teacher-forcing decay up to 0.5, and gradual unfreezing of RoBERTa layers with the schema described in section 4.2.

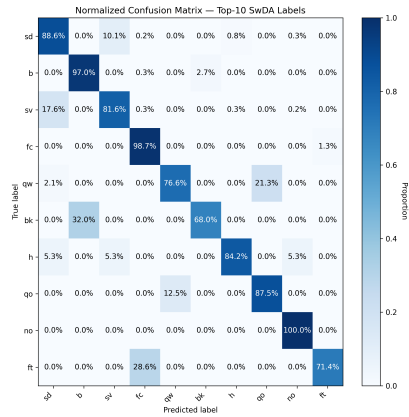
Even though the model achieved a boost in performance when compared to the retrained baseline, in line with our initial goal, we were expecting a bigger gap in performance given the more structured decoding approach.

The reason for such minimal performance boost could be that the embeddings produced by RoBERTa are not sufficient to allow huge accuracy improvements. For this reason a possible future implementation could be to test out different pretrained models for the utterance embeddings, specifically dialogue focused ones such as the Dialogue Transformer Language Model (Nguyen et al. [2023]), whose conversational pretraining objectives may yield richer utterance embeddings than our general purpose RoBERTa backbone. Other options to widen the performance gap could be adopting better fine tuning strategies, such as applying less aggressive sequence level risk minimization losses to encourage the decoder to explore and select more globally optimal label sequences. To instead close the remaining distance to state of the art results ($\sim 7\%$) on SwDA, we could investigate alternative preprocessing pipelines (including refined disfluency stripping, interruption merging, and label aware oversampling) alongside cost sensitive and data augmentation techniques to mitigate the corpus’s severe class imbalance.

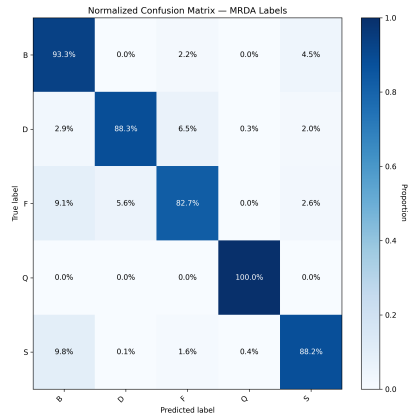
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A Confusion matrices for Baseline model

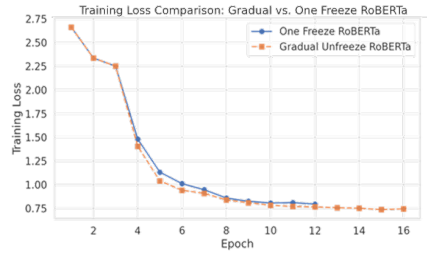


(a) He et al. baseline confusion matrix for top-10 SwDA labels

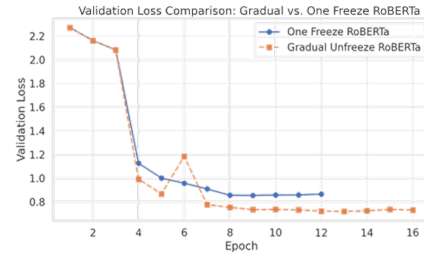


(b) He et al. baseline confusion matrix for MRDA labels

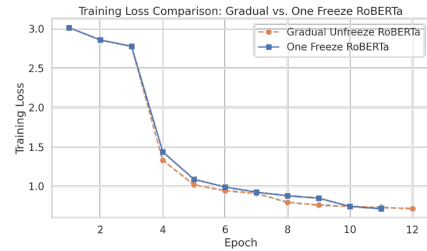
B One-Freeze vs Gradual Unfreezing



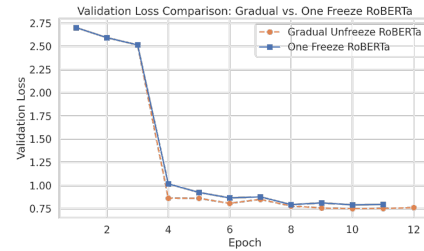
(a) Training loss comparison between One Freeze and Gradual Unfreeze for the Bi-GRUBERTa model



(b) Validation loss comparison between One Freeze and Gradual Unfreeze for the Bi-GRUBERTa model



(c) Training loss comparison between One Freeze and Gradual Unfreeze for He et al. baseline model



(d) Validation loss comparison between One Freeze and Gradual Unfreeze for He et al. baseline model